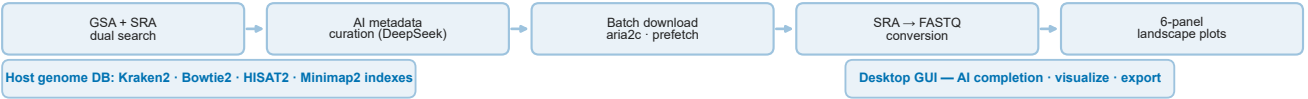


(a) Public Data Acquisition & Curation

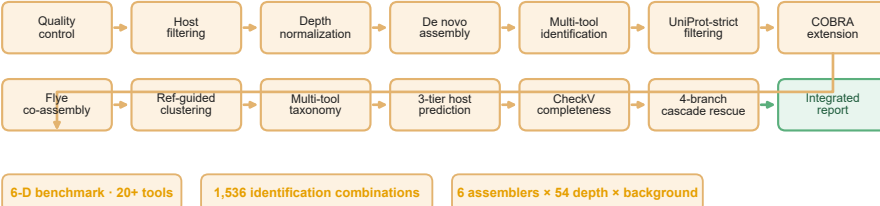
public_metadata_pipeline



(b) Benchmark-Driven Discovery Pipeline

virome_discovery_pipeline

13 stages · every default parameter benchmark-validated

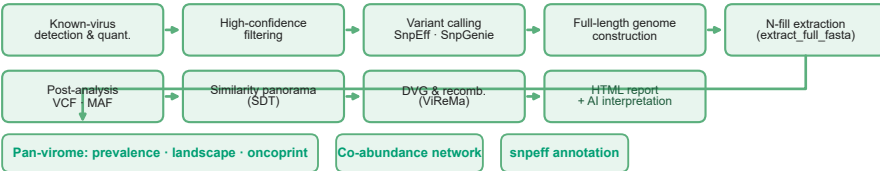


4-branch cascade rescue

- A CheckV ≥ 90% complete → pass
- B Virseqimprover extension
- C BLASTN → PlantVirusDB rebuild: bwa-mem2 + viral_consensus / ragtag
- D Genus-length rescue (fallback)

(c) Rapid Known-Virus Analysis

virome_analysis_pipeline

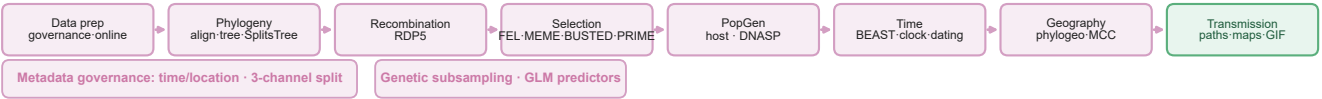


OmniVirusAssembler (stage 'full')

- 12-step refinement · SHIVER-like fusion skeleton
- Reads-level iterative polishing — single-base precision
- Dual-engine gap closing: gmcloser + abyss-sealer
- Circularization detection → [Circular=True]

(d) Phylodynamics & Evolution

virome_phylo_pipeline



(e) Semi-automatic GenBank Submission

virome_submission_pipeline



Desktop GUI (PySide6)

(f) CROSS-CUTTING DESIGN STRATEGIES

Data Reduction

99.8% *host-filtering efficiency*

- Reference-guided pre-clustering
- Host-based sequence partitioning

Quality Enhancement

4-branch *cascade genome rescue*

- COBRA resolves de Bruijn breakpoints
- Flye co-assembly merges cross-sample
- BLASTN → PlantVirusDB · viral_consensus / ragtag
- Genus-length rescue as final fallback

Reliability Enhancement

F1 = 0.942 *identification (UniProt-strict)*

- UniProt-strict ID ≈ 5-tool ensemble
- Family accuracy 94.1% (+72.3 pp)
- 3-tier host + plant-specific filter

(g) SHARED INFRASTRUCTURE

Tool Library

30+ wrapped: Kraken2 · Bowtie2 · snpEff · VirHunter ViReMa · BEAST · RDP5 · HyPhy · CheckV

Reference Databases

ICTV VMR taxonomy · PlantVirusDB host genomes · UniProt / VP

Metadata Governance

Time/location parsing & normalization three-channel split

Reproducible Environment

pixi-pinned environments stage checkpoints · one-command reruns